Topic 15: Transition Metals

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include the stepwise reduction of vanadium(V) to vanadium(II), investigating the reactions of copper(II) ions or chromium(III) ions, using sodium hydroxide and ammonia solution to identify transition metal ions, investigating autocatalysis, preparing a complex transition metal salt.

Mathematical skills that could be developed in this topic include investigating the geometry of different transition metal complexes.

Within this topic, students can consider the model for the filling of electron orbitals encountered in Topic 1, and see how limitations in that model indicate the need for more sophisticated explanations. They can also appreciate that catalyst research is a frontier area, and one which provides an opportunity to show how the scientific community reports and validates new knowledge.

Students should:
Topic 15A: Principles of transition metal chemistry
1. be able to deduce the electronic configurations of atoms and ions of the *d*-block elements of period 4 (Sc–Zn), given the atomic number and charge (if any)
2. know that transition metals are *d*-block elements that form one or more stable ions with incompletely-filled *d*-orbitals
3. understand why transition metals show variable oxidation number
4. know what is meant by the term 'ligand'
5. understand that dative (coordinate) bonding is involved in the formation of complex ions
6. know that a complex ion is a central metal ion surrounded by ligands
7. know that transition metals form coloured ions in solution
8. understand that the colour of aqueous ions, and other complex ions, results from the splitting of the energy levels of the *d*-orbitals by ligands
9. understand why there is a lack of colour in some aqueous ions and other complex ions
10. understand that colour changes in transition metal ions may arise as a result of changes in: i oxidation number ii ligand iii coordination number
11. understand the meaning of the term 'coordination number'
12. understand why H_2O, OH^- and NH_3 act as monodentate ligands
13. understand why complexes with six-fold coordination have an octahedral shape, such as those formed by metal ions with H_2O, OH^- and NH_3 as ligands

Students should:
14. know that transition metal ions may form tetrahedral complexes with relatively large ligands such as Cl^-
15. know that square planar complexes are also formed by transition metal ions and that *cis*-platin is an example of such a complex
16. understand why *cis*-platin used in cancer treatment is supplied as a single isomer and not in a mixture with the *trans* form
17. be able to identify bidentate ligands, such as $NH_2CH_2CH_2NH_2$ and multidentate ligands, such as $EDTA^{4-}$
18. know that haemoglobin is an iron(II) complex containing a multidentate ligand *The structure of the haem group will not be assessed.*
19. know that a ligand exchange reaction occurs when an oxygen molecule bound to haemoglobin is replaced by a carbon monoxide molecule